Say it, then write it: choose an equation

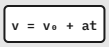
Language goal: I can **explain** why I chose an equation.



given
dato · dado
a number the problem tells you



unknown
incógnita · incógnita
the value you must find



equation
ecuación · equação
a rule that links the values



constant acceleration
aceleración constante · aceleração constante
v changes the same amount each second

English · Spanish · Portuguese. My home language: _____

1 · READ AND FILL · WORD BANK: given · unknown · equation · time · steady

First I list each ____ value. Then I name the ____ I must find. I choose the ____ that uses only what I have. If no ____ is given, I use $v^2 = v_0^2 + 2a\Delta x$. These equations work only when acceleration is ____.

2 · SAY IT TO A PARTNER, THEN WRITE IT



I know ____, ____, and ____. I need ____ .
I choose ____ because it does not need ____ .

3 · SOLVE WITH THE STEPS



0 → ? m/s
a = 3 m/s² · 4 s

A car starts at 0 m/s and speeds up at 3 m/s² for 4 s. How fast is it going?

STEP 1 · LIST
v_0 = ____ · a = ____ · t = ____

STEP 2 · CHOOSE + SOLVE
$v = v_0 + at$ = ____ + ____ × ____

STEP 3 · SAY IT
"After 4 s the car moves ____ m/s."

4 · EXPLAIN IT · CHOOSE A FRAME

☐ I chose this equation because ____ .
☐ I did not use ____ because the problem does not give ____ .
☐ First I listed ____, then I found ____ .

I said it ☐ · I wrote it ☐

R5 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Unlock the code: constant acceleration

08 C

DIRECTIONS · Solve each station. Copy the letter into the lock.

STATION 1

v_0 = 1 m/s, a = 2 m/s², t = 2 s. Final v?

(S) 5 m/s (T) 4 m/s (A) 3 m/s

STATION 2

From rest, a = 3 m/s² for 4 s. $\Delta x = \frac{1}{2}at^2$?

(P) 24 m (E) 12 m (K) 48 m

STATION 3

0 → 8 m/s in 4 s, steady. Δx using average v?

(E) 16 m (O) 32 m

STATION 4

20 m/s, a = −4 m/s², to a stop. Δx ($v^2 = v_0^2 + 2a\Delta x$)?

(E) 50 m (I) 100 m (U) 5 m

STATION 5

Steady +2 m/s for 3 s. Δx?

(D) 6 m (N) 5 m

STATION 6

Which equation needs NO time?

(S) $v^2 = v_0^2 + 2a\Delta x$ (R) $v = v_0 + at$

LOCK 1 2 3 4 5 6

Hint: a word for velocity × time… or what a car does at a green light.

R5 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

ANTOCCI ENGINEERING · DWG U1-08 · R5

BRAKE ZONE: SCHOOL CROSSWALK

Design spec: cars approach at 15 m/s. The crosswalk is 25 m past the warning sign.

CALC 1 · NEEDED BRAKING

Stop in 25 m: use $v^2 = v_0^2 + 2a\Delta x$. $a =$ ____ m/s²

CALC 2 · STOPPING TIME

$v = v_0 + at \rightarrow t =$ ____ s

DRAWING · SKETCH THE v–t GRAPH FOR THE STOP (label the area)

CALC 3 · CHECK WITH AREA

$\Delta x = \frac{1}{2}(v_0 + v)t =$ ____ m. Does it match 25 m?

CALC 4 · WET ROAD

If the best braking is only −3 m/s², how far is the stop? Safe?

ENGINEER'S NOTE TO THE CITY

Where should the warning sign go? Justify with your numbers.

R5 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Four steps, every equation

08 M

DIRECTIONS · Work each problem through all four boxes.

PROBLEM 1 · A train starts from rest and accelerates at 0.50 m/s² until it reaches 80 m/s. How long does it take?

1 READ

v_0 ____ · v ____ · a ____ · find ____

2 DRAW

v–t sketch

3 SOLVE

choose: $v = v_0 + at$

4 WRITE

It takes ____ s. I chose this equation because ____ .

PROBLEM 2 · A car moving 20 m/s brakes at −4 m/s². How far does it travel before stopping?

1 READ

v_0 ____ · v ____ · a ____ · no time given

2 DRAW

sketch the stop

3 SOLVE

$v^2 = v_0^2 + 2a\Delta x$

4 WRITE

The car stops in ____ m. Negative a, positive Δx because ____ .

Hardest box: □ Read □ Draw □ Solve □ Write

R5 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Talk it through: choosing an equation

Language goal: I can **justify** a choice.

DIRECTIONS · Read your lines aloud. Listen, then write. Switch roles each round.

ROUND 1 · 0 → 12 m/s AT 4 m/s². HOW LONG?



0 → 12 m/s
a = 4 m/s²

A ASKS "Which equation should we use?"

B SAYS "We know v0, v and a. We need ____, so I choose ____."

A CHECKS "So t = ____ s. I agree / disagree because ____ ."

ROUND 2 · SWITCH · BRAKING FROM 20 m/s AT −4 m/s²



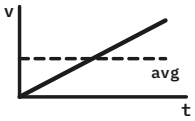
20 m/s → 0
a = −4 m/s²

B ASKS "How far until it stops? There's no time!"

A SAYS "No time means I use ____. Δx = ____ m."

B ADDS "I want to add that a is negative but Δx is ____ because ____."

ROUND 3 · BOTH · CHECK WITH AVERAGE VELOCITY



A "For steady a, average v = (v0 + v) ÷ 2 = ____."

B "Then Δx = average v × t = ____. It matches / doesn't match."

LISTENING LOG · MY PARTNER SAID…

NOW I WRITE IT MYSELF

I choose an equation by...

Stems: I agree because… · Can you say that again? · What does ___ mean? · I want to add…

R5 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Which equation fits?

08 N

DIRECTIONS · Read each problem card. Write its letter under the equation you'd use. (Steady a for all.)

A Find final v after 3 s at 2 m/s².	**B** How far in 4 s from rest at 3 m/s²?	**C** Braking distance from 20 m/s at −4 m/s².	**D** How long to reach 12 m/s from 0 at 4 m/s²?
E Speed after rolling 9 m from rest at 2 m/s².	**F** Displacement in 5 s starting at 2 m/s, a = 1.	**G** Launch at 30 m/s, a = −10: velocity at 2 s?	**H** Stopping from 15 m/s in 5 m: find a.
I How far while 0 → 8 m/s in 4 s?	**J** Time to stop from 10 m/s at −2 m/s².	**K** Top of a toss: v = 0. Height if v0 = 20 m/s?	**L** Car at 5 m/s, a = 3 for 2 s: final v?

$v = v_0 + at$	$\Delta x = v_0 t + \frac{1}{2}at^2$	$v^2 = v_0^2 + 2a\Delta x$
need v or t	time is given	no time given

EXPLAIN ONE TRICKY CARD

Card ____ belongs where it does because _____

R5 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Comic: 3 m/s faster every second

08 CS

YOUR READER

A younger student who thinks "faster every second" means the same distance every second.

Stick figures are fine. The physics gets checked.

MUST-HAVES · check it off, then color its circle

- ☐ ◌ **GREEN** — The car in every panel (it is speeding up)
- ☐ ◌ **PURPLE** — Velocity arrows that grow each panel, labeled 0, 3, 6 m/s
- ☐ ◌ **BLUE** — Position dots that spread farther apart

1 · t = 0 s *A car at rest at the start line.*

CAPTION _____

2 · t = 1 s *The car is now going 3 m/s.*

CAPTION _____

3 · t = 2 s *The car is now going 6 m/s.*

CAPTION _____

EXPLAIN IT · In 1–2 sentences, what physics does your comic show? Use: **acceleration · velocity · constant · m/s²**

R5 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Doodle Notes · constant acceleration

I can draw a change and explain what my marks mean.

Name: _____ My home language: _____

Constant acceleration · aceleración constante (ES) · aceleração constante (PT)



TODAY'S CASE

A car starts from rest and gains 3 m/s each second on a straight road.

01 · SKETCH AND LABEL

Draw purple velocity arrows at t = 0, 1 and 2 s; label the zero velocity without giving it a length.

02 · CHANGE THE PICTURE

Sketch blue position dots for those times. Explain why equal changes in velocity do not give equal distances in equal times.

Annotate your sketches. Circle the mark that best shows constant acceleration. Add a label in English or your home language.

SAY IT TO A PARTNER · THEN WRITE

The arrows change by ___ each second, while the gaps …

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Frayer Model · constant acceleration

I can explain a term by comparing an example with a nonexample.

Name: _____ My home language: _____

Constant acceleration · aceleración constante (ES) · aceleração constante (PT)

A · A car gains 3 m/s in each successive second.

B · A car moves at +3 m/s the entire time and therefore gains 3 m/s each second.

01 · DEFINITION

What does constant acceleration mean in your own words? Use the notes or start in your home language.

02 · CHARACTERISTICS

What must be true? Label a small sketch to show one feature.

03 · EXAMPLE

Choose A or B as your example. Sketch it and explain which feature makes it fit.

04 · NONEXAMPLE

Explain why the other card does not fit. Change one detail so that it would.

SAY IT TO A PARTNER · THEN WRITE

I chose card ___ as my example because …

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Picture Dictionary · constant acceleration

I can connect each word to a picture and use it to explain this case.

Name: _____ My home language: _____



given

dato · dado

a number the problem tells you

My word or label: _____

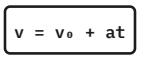


unknown

incógnita · incógnita

the value you must find

My word or label: _____



equation

ecuación · equação

a rule that links the values

My word or label: _____

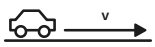


constant acceleration

aceleración constante · aceleração constante

v changes the same amount each second

My word or label: _____

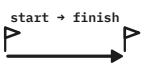


velocity

velocidad · velocidade

speed with its direction

My word or label: _____



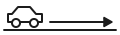
displacement

desplazamiento · deslocamento

change from the start to the finish

My word or label: _____

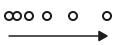
MATCH · WRITE THE ENGLISH WORD UNDER EACH PICTURE



_____



_____



_____

Use two words to explain the case: A car starts from rest and gains 3 m/s each second on a straight road.

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else